EEE 452 - Homework 2

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Question 1

a)

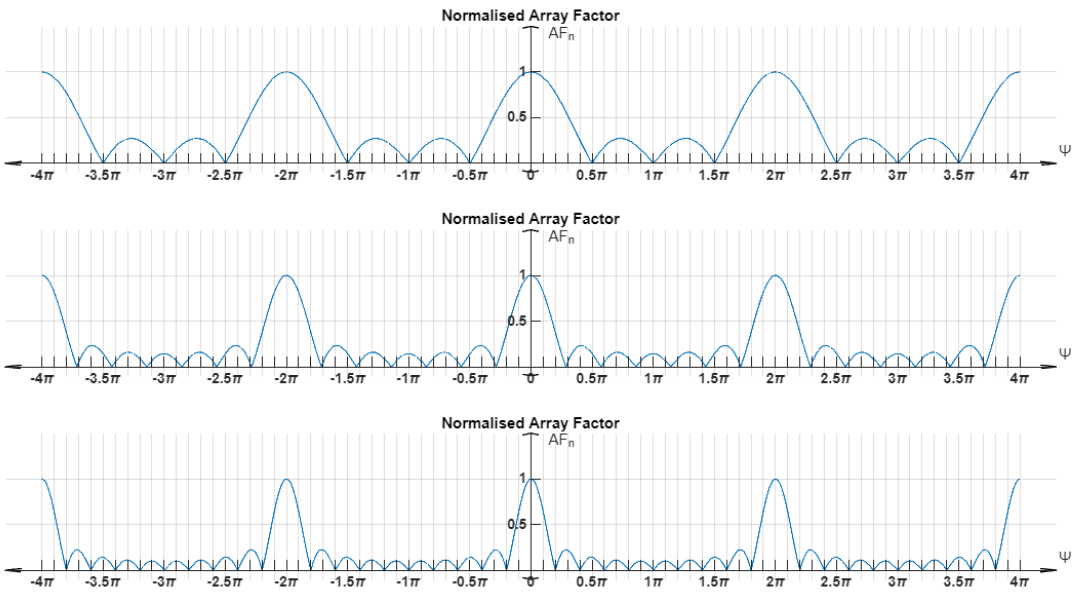

Figure 1.1.1: Normalized Array Factor for N=4, 7, 10

We can see the normalised array factors for linear arrays with 4, 7, and 10 elements respectively.

N	4	7	10	Formulas
Number of Nulls	3	6	9	N-1
Width of the Main Lobe	π	0.57π	0.4π	$\frac{4\pi}{N}$
Width of the Side Lobes	0.5π	0.28π	0.2π	$\frac{2\pi}{N}$
Number of Sidelobes	2	5	8	N-2
Null Locations	0.5π, π, 1.5π	0.28π, 0.57π, 0.85π, 1.14π, 1.42π, 1.71π	0.2π, 0.4π, 0.6π, 0.8π, π, 1.2π, 1.4π, 1.6π, 1.8π	$\frac{2\pi}{N}n, n \in Z, n \notin kN, k \in Z$

Table 1.1.1: Table of Parameters of the Array Factors

The values obtained from the plots match perfectly with their formulas.

b)

We expect the sidelobes to decay really fast and the plot will resemble a periodic impulse function with impulses at 2π.

c)

For this section we have 27 plots with the combinations of:

$$N = 4, 7, 10 \qquad d = 0.5\lambda, \lambda, 1.5\lambda \qquad a = 0, 0.5\pi, \pi$$

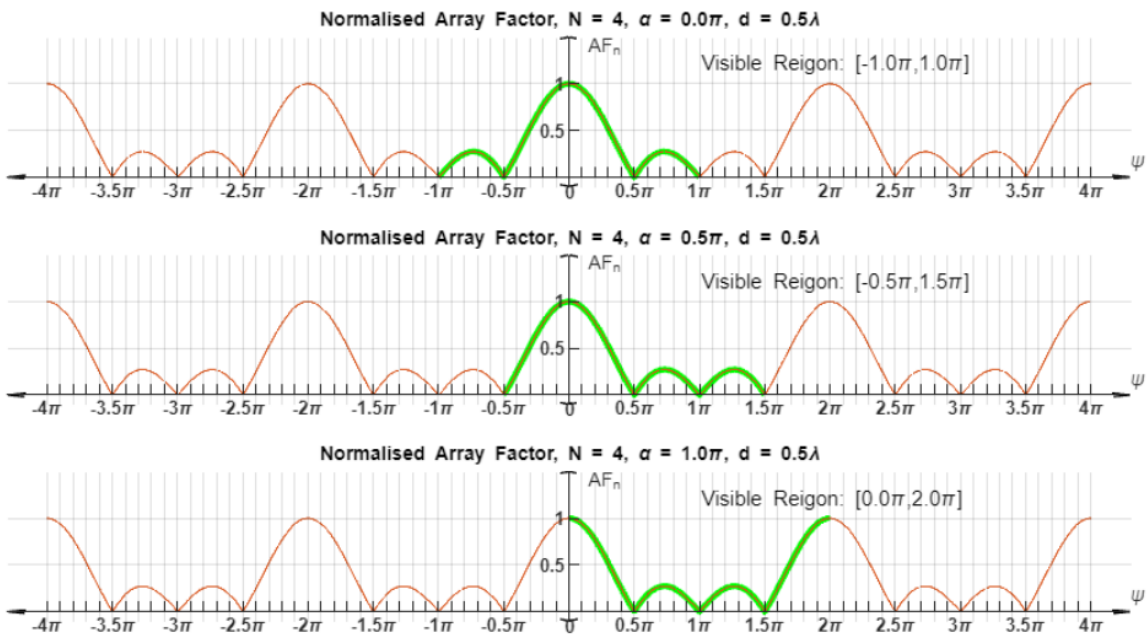

Figure 1.3.1: Various a Values for N=4, $d = 0.5\lambda$

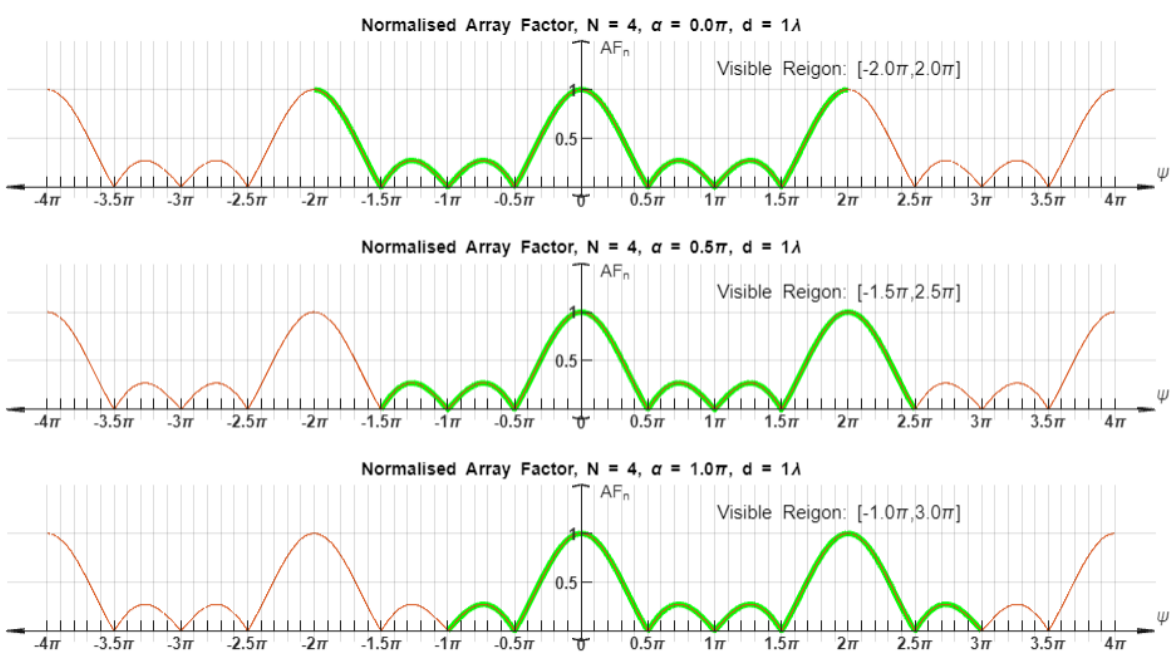

Figure 1.3.2: Various a Values for N=4, $d = 1\lambda$

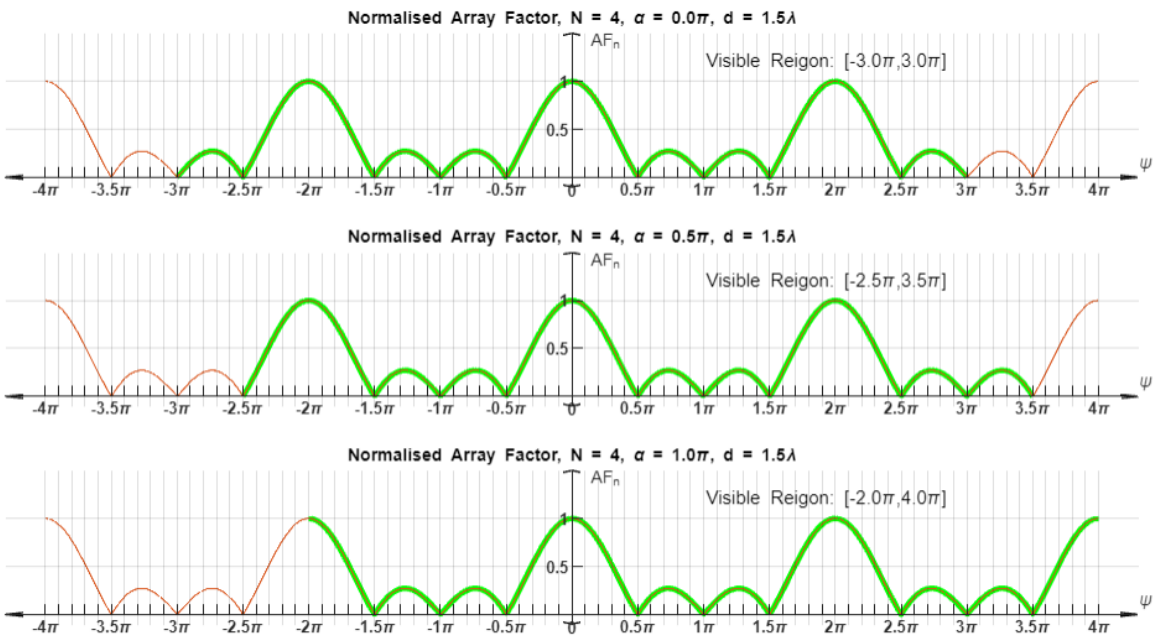

Figure 1.3.3: Various a Values for N=4, $d = 1.5\lambda$

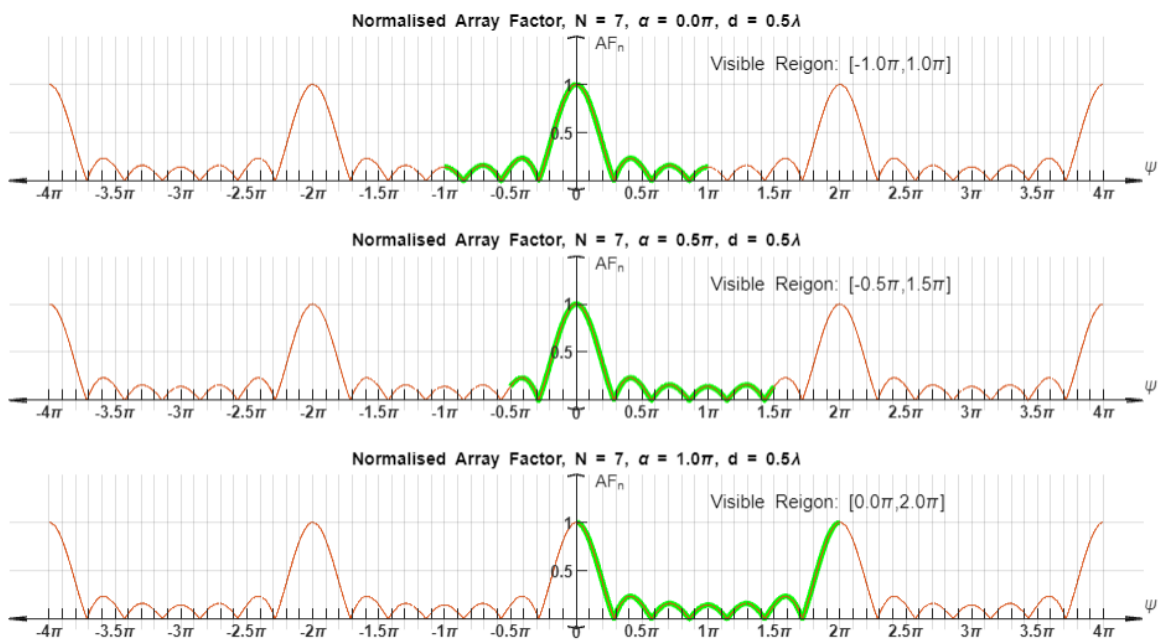

Figure 1.3.4: Various a Values for N=7, $d = 0.5\lambda$

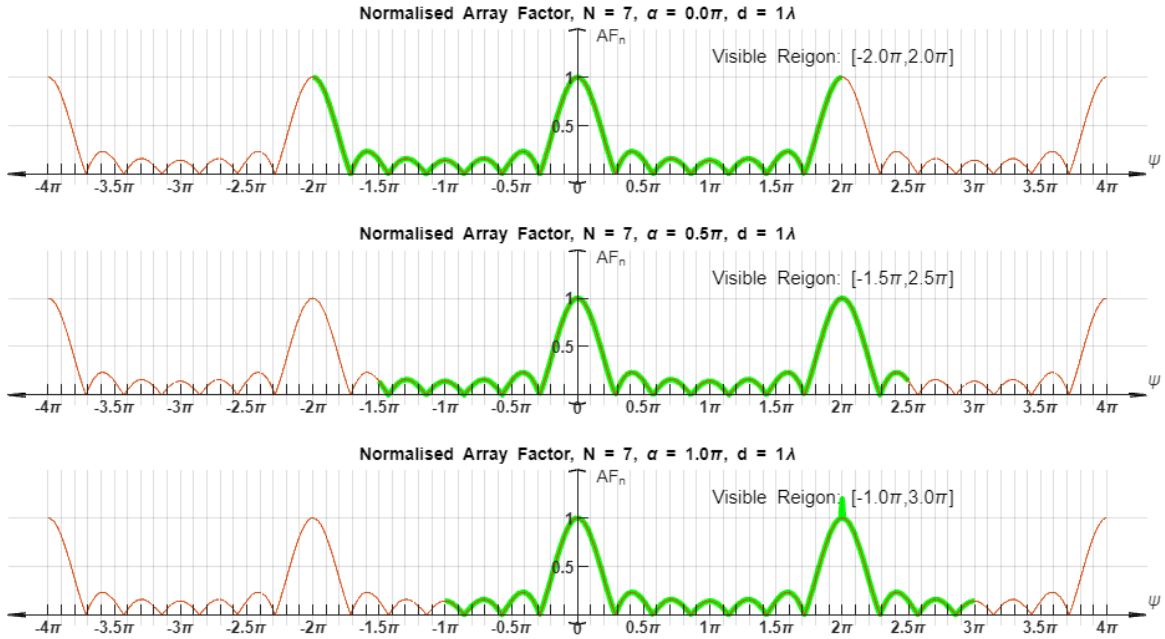

Figure 1.3.5: Various a Values for N=7, $d = 1\lambda$

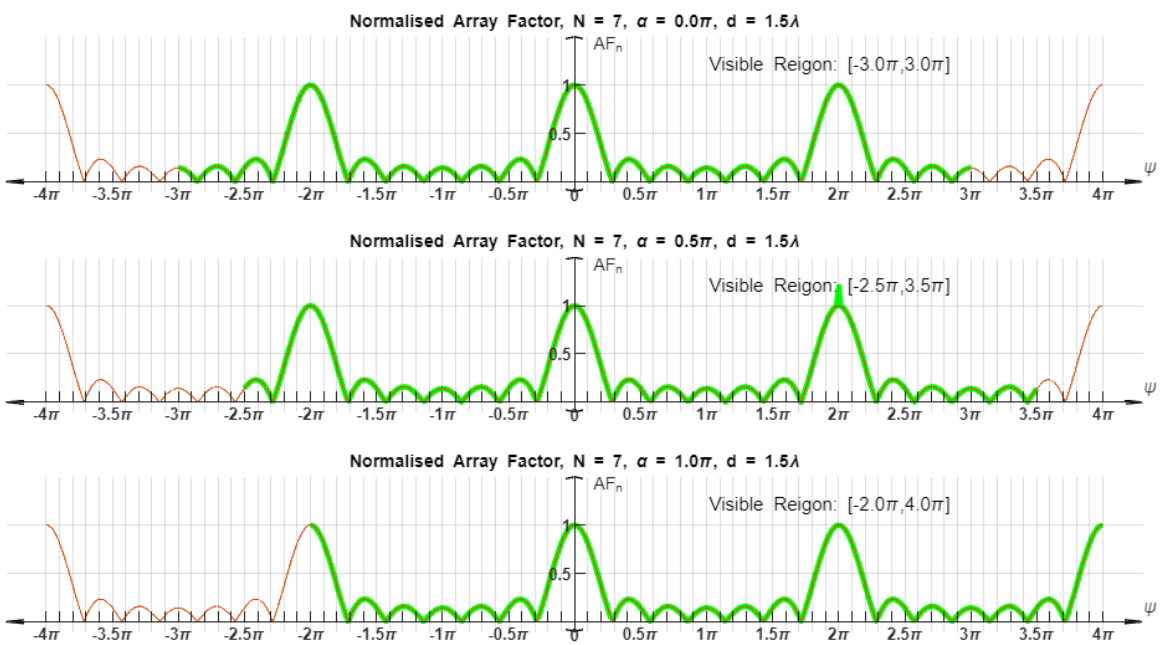

Figure 1.3.6: Various a Values for N=7, $d = 1.5\lambda$

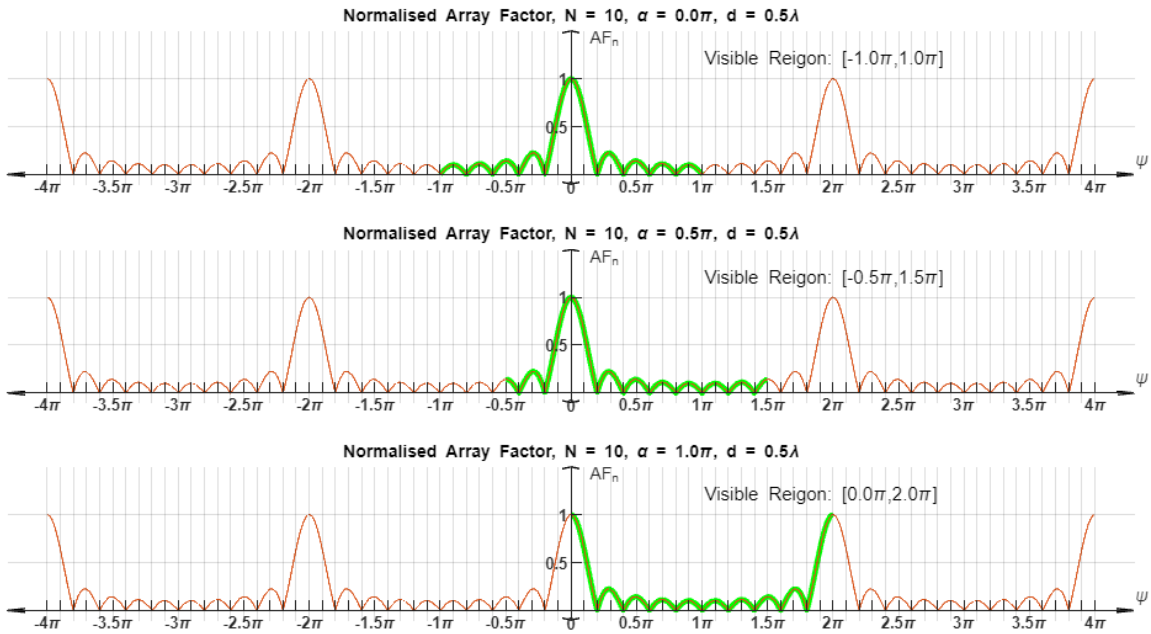

Figure 1.3.7: Various a Values for N=10, $d = 0.5\lambda$

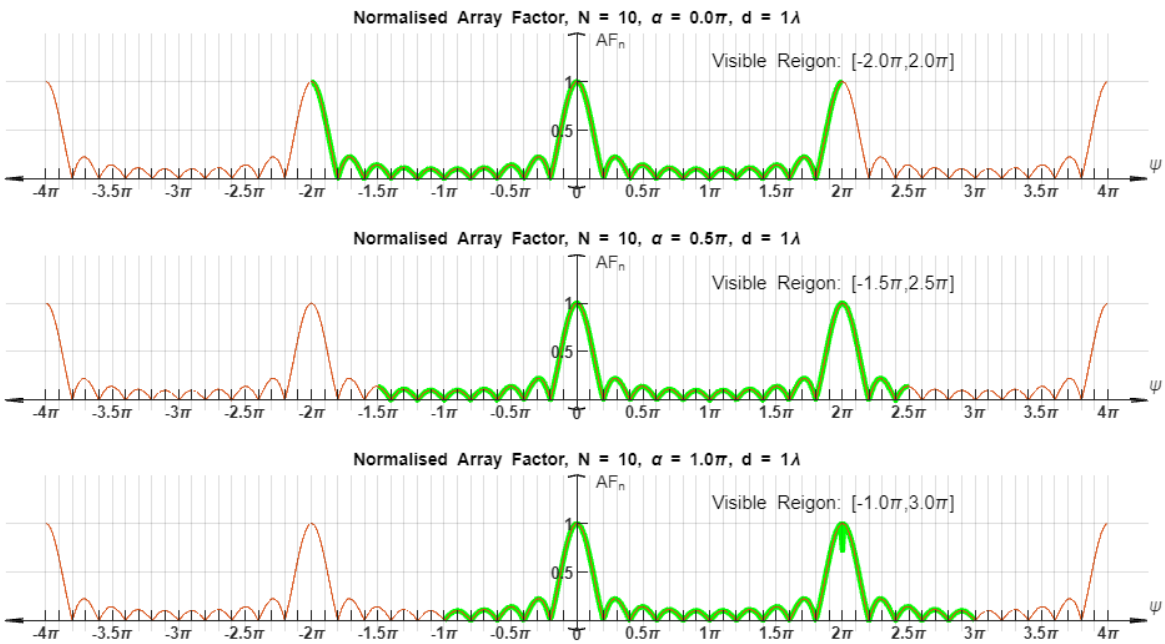

Figure 1.3.8: Various a Values for N=10, $d = 1\lambda$

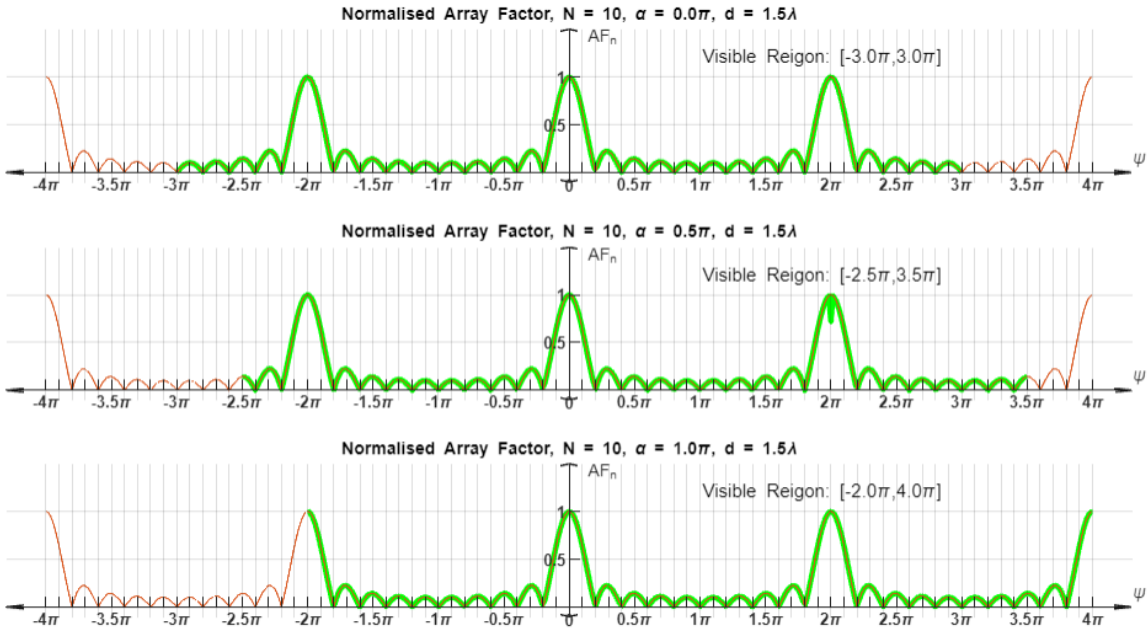

Figure 1.3.9: Various a Values for N=10, $d = 1.5\lambda$

d)

The theta value for maximum (mainlobe) is given by the following formula:

$$\theta_m = \cos^{-1}\left(-a\frac{\lambda}{2\pi d}\right) \tag{1.4.1}$$

Increasing a decreases this value, and increasing d brings it closer to 0.

e)

For there not to be any grating lobes, the value of ψ must stay between $[-2\pi, 2\pi]$. For a given as $-kd\cos(\theta)$,

$$-2\pi < kd\left[\cos(\theta) - \cos(\theta_0)\right] < 2\pi \quad \Rightarrow -1 < \frac{d}{\lambda}\left[\cos(\theta) - \cos(\theta_0)\right] < 1 \tag{1.5.1}$$

This can be split into two inequalities considering $-1 < \cos(\theta) < 1$:

$$\boxed{\frac{d}{\lambda}(1 - \cos(\theta_0)) < 1}, \quad -1 < \frac{d}{\lambda}(1 - \cos(\theta_0)) \Rightarrow \boxed{\frac{d}{\lambda}(1 + \cos(\theta_0)) < 1} \tag{1.5.2}$$

We can combine these two inequalities into a single one by taking the absolute value of the $\cos(\theta_0)$ term.

$$\frac{d}{\lambda}(1 + |\cos(\theta_0)|) < 1 \quad \Rightarrow \quad \boxed{d < \frac{\lambda}{1 + |\cos(\theta_0)|}} \tag{1.5.3}$$

$$d < \frac{\lambda}{1 + |\cos(\theta_0)|} \iff -2\pi < kd\left[\cos(\theta) - \cos(\theta_0)\right] < 2\pi \iff \text{No Grating Lobes} \tag{1.5.4}$$

f)

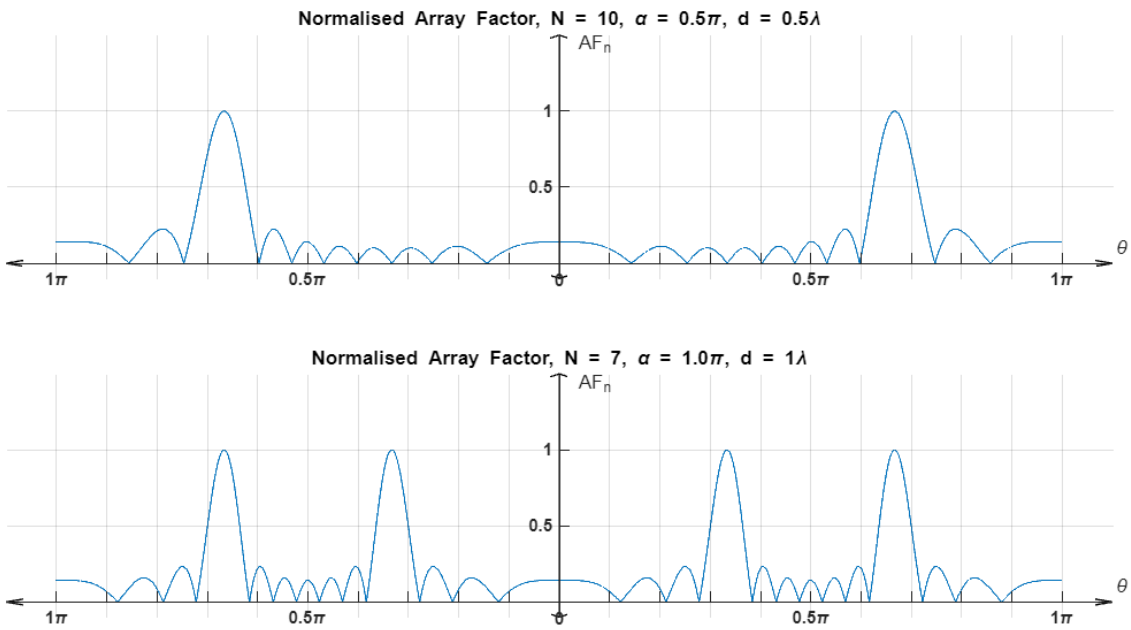

Figure 1.6.1: Array Factor as a Function of θ for $N = 10, a = 0.5\pi, d = 0.5\lambda$ and $N = 7, a = \pi, d = \lambda$

The visible range is mapped tp $[-\pi, \pi]$ with a non-linear relation of $\cos(\theta)$. This results in the visible reigon to get stretched towards $0, \pm\pi$ and squeezed around $\pm 0.5\pi$.

Question 2

a)

In this section we will find the array pattern of a z-directed linear array consisting of 5 z-directed infinitesimal dipoles.

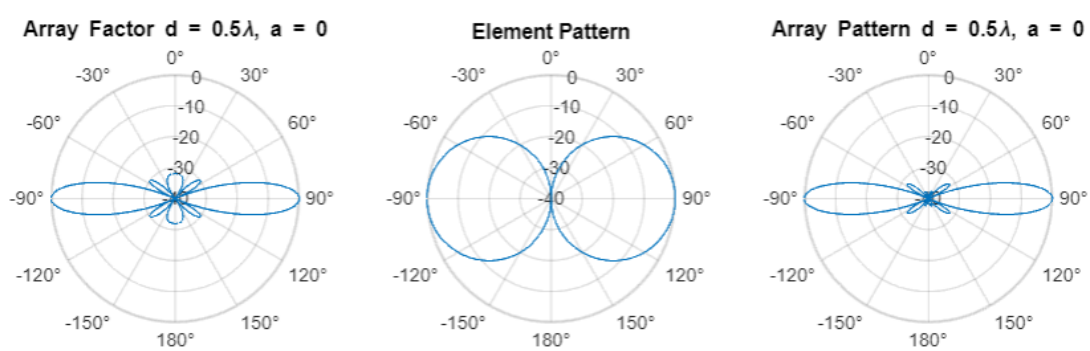

Figure 2.1.1: Array Factor, Element Pattern and the Array Pattern

Due to the principal of pattern multiplication, the pattern of a single antenna was multiplied with the array factor. Both have their mainlobes at $\theta = \pi/2$, hence the resulting pattern has a more directive mainlobe at $\theta = 0.5\pi$.

b)

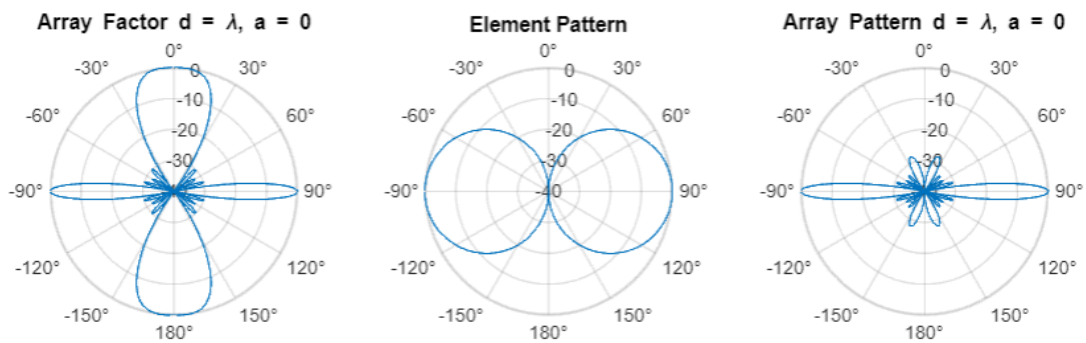

Figure 2.2.1: Array Factor, Element Pattern and the Array Pattern

We can see that the sidelobes toward $\theta = 0, \pi$ increased in amplitude due to the grating lobes. This will change the gain of the antenna as the power radiated in the sidelobes also increase.

c)

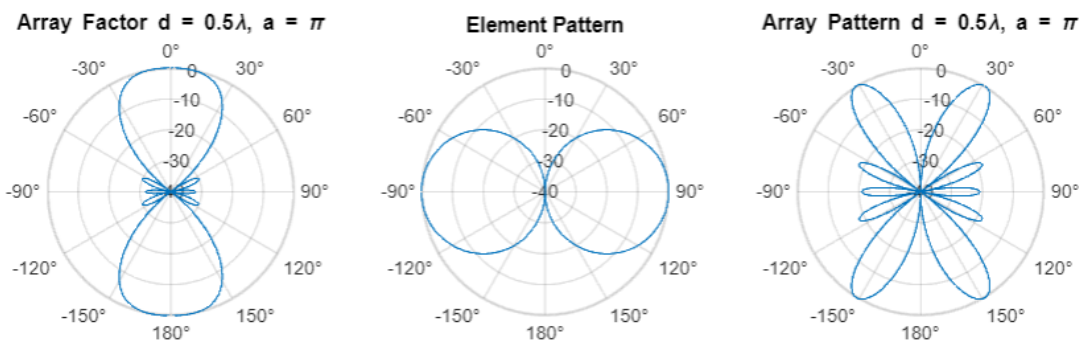

Figure 2.3.1: Array Factor, Element Pattern and the Array Pattern

The array factor's main beam coincides with a zero of the element pattern hence effectively cutting the main beam in half. This decreases the mainlobe's amplitude significantly and this results in a really high side lobe level.

d)

The required a for these mainlobe angles can be calculated by the formula:

$$\theta_m = \cos^{-1}\left(-a\frac{\lambda}{2\pi d}\right) \tag{2.4.1}$$

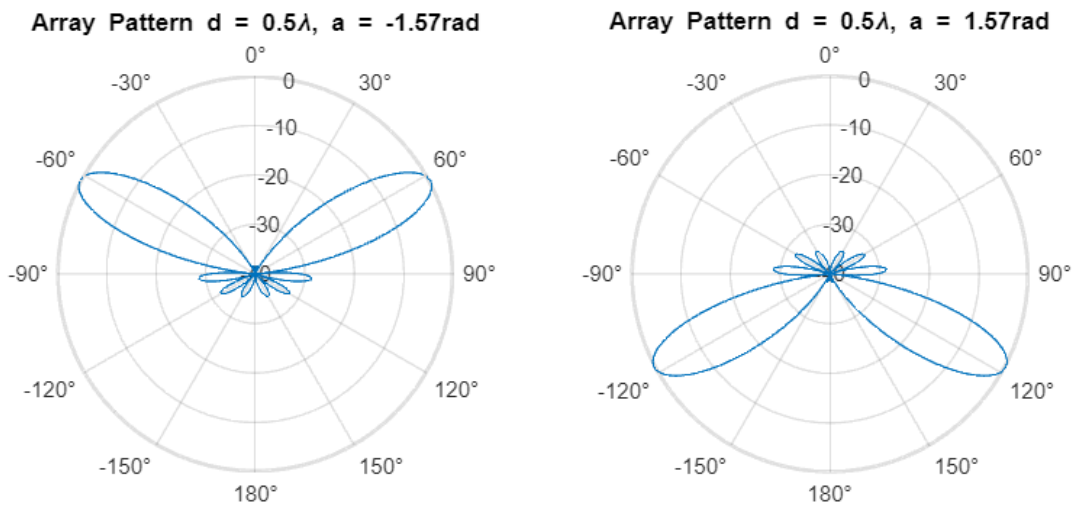

Figure 2.4.1: Phase Steered Patterns at $\pi/3, 2\pi/3$

1-a)

```
x = -4:0.1:4;
labels = get_labels(x);
i = 1;
for N = [4 7 10]
    psi = linspace(-4*pi,4*pi,10000);
    AF_n = abs(1/N .* sin(N*psi/2) ./ sin(psi/2));
    subplot(3,1,i)
    fineplot(psi,AF_n,"Normalised Array Factor","\Psi","AF_n",[-4.3*pi 4.3*pi],
[-0.1 1.5],"off",[1200 600],"","-");
    i = i + 1;
    xticks(x.*pi)
    xticklabels(labels)
    xtickangle(0)
    grid on
end
```

1-c)

```
x = -4:0.1:4;
labels = get_labels(x);
theta = linspace(-pi,pi,1000);
for N = [4 7 10]
    for d = [0.5 1 1.5]
        i = 1;
        figure;
        for a = [0 pi/2 pi]
            psi = linspace(-4*pi,4*pi,10000);
            AF_n = abs(1/N .* sin(N*psi/2) ./ sin(psi/2));
            psi2 = 2*pi*d*cos(theta) + a;
            AF_n2 = abs(1/N .* sin(N*psi2/2) ./ sin(psi2/2));
            subplot(3,1,i)
            plot(psi2,AF_n2,"color",[0 1 0 0.1],"LineWidth",3)
            hold on
            fineplot(psi,AF_n,strcat("Normalised Array Factor, N
= ",num2str(N),", \alpha = ",num2str(a/pi,"%.1f"),"\pi, d =
",num2str(d),"\lambda"),"\theta","AF_n",[-4.3*pi 4.3*pi],[-0.1 1.5],"off",[1200
600],"","-");
            text(pi,1.2,strcat("Visible Reigon: [",num2str(min(psi2)/
pi,"%.1f"),"\pi,",num2str(max(psi2)/pi,"%.1f"),"\pi]"))
            xticks(x*pi)
            xticklabels(labels)
            xtickangle(0)
            i = i + 1;
            grid on
        end
    end
end
```

1-f)

```
x = -1:0.1:1;
labels = get_labels(abs(x));
theta = linspace(-pi,pi+0.001,10000); %+0.001 exists to avoid a discontunity at 0
d = 0.5;
a = pi/2;
N = 10;
for i = 1:2
    psi = 2*pi*d*cos(theta) + a;
    AF_n = abs(1/N .* sin(N*psi/2) ./ sin(psi/2));
    subplot(2,1,i)
    fineplot(theta,AF_n,strcat("Normalised Array Factor, N =
",num2str(N),", \alpha = ",num2str(a/pi,"%.1f"),"\pi, d =
",num2str(d),"\lambda"),"\theta","AF_n",[-1.1*pi 1.1*pi],[-0.1 1.5],"off",[1200
600],"","-");
    xticks(x*pi)
    xticklabels(labels)
    xtickangle(0)
    grid on
    d = 1;
    a = pi;
    N = 7;
end
```

2-a)

```
clf;
theta = linspace(-pi,pi,1000);
psi = 2*pi*0.5*cos(theta) + 0;
AF = 1/5 .* sin(5*psi/2) ./ sin(psi/2);
EF = sin(theta);
subplot(1,3,1)
pattern_plot(theta,AF,"-")
title("Array Factor d = 0.5\lambda, a = 0")
subplot(1,3,2)
pattern_plot(theta,EF,"-")
title("Element Pattern")
subplot(1,3,3)
pattern_plot(theta,EF.*AF,"-")
title("Array Pattern d = 0.5\lambda, a = 0")
set(gcf,'position',[1 , 1 , 1200 , 600])
```

2-b)

```
clf;
theta = linspace(-pi,pi,1000);
psi = 2*pi*1*cos(theta) + 0;
AF = 1/5 .* sin(5*psi/2) ./ sin(psi/2);
EF = sin(theta);
subplot(1,3,1)
pattern_plot(theta,AF,"-")
title("Array Factor d = \lambda, a = 0")
subplot(1,3,2)
```

```
pattern_plot(theta,EF,"-")
title("Element Pattern")
subplot(1,3,3)
pattern_plot(theta,EF.*AF,"-")
title("Array Pattern d = \lambda, a = 0")
set(gcf,'position',[1 , 1 , 1200 , 600])
```

2-c)

```
clf;
theta = linspace(-pi,pi,1000);
psi = 2*pi*0.5*cos(theta) + pi;
AF = 1/5 .* sin(5*psi/2) ./ sin(psi/2);
EF = sin(theta);
subplot(1,3,1)
pattern_plot(theta,AF,"-")
title("Array Factor d = 0.5\lambda, a = \pi")
subplot(1,3,2)
pattern_plot(theta,EF,"-")
title("Element Pattern")
subplot(1,3,3)
pattern_plot(theta,EF.*AF,"-")
title("Array Pattern d = 0.5\lambda, a = \pi")
set(gcf,'position',[1 , 1 , 1200 , 600])
```

2-d)

```
EF = sin(theta);
clf;
for tm = 1:2
    theta = linspace(-pi,pi,1000);
    a = -cos(tm*pi/3)*2*pi*0.5;
    psi = 2*pi*0.5*cos(theta) + a;
    AF = 1/5 .* sin(5*psi/2) ./ sin(psi/2);
    subplot(1,2,tm)
    pattern_plot(theta,EF.*AF,"-")
    title(strcat("Array Pattern d = 0.5\lambda, a = ",num2str(a,"%.2f"),"rad"))
end
set(gcf,'position',[1 , 1 , 800 , 400])
```

Functions

```
function labels = get_labels(x)
    labels = strings(size(x));
    for k = 1:length(x)
        if mod(x(k),0.5) == 0
            if x(k) == 0
                labels(k) = "0";
            else
                labels(k) = sprintf('%g\\pi', x(k));
            end
        else
```

```
            labels(k) = "";
        end
    end
end

function y = normalise(x)
    y = x/max(x);
end

function y = trunc(x) % rounds everything below -40dB up to -40dB. Maps -40,0 to
0,1
    y = ((x+40).*(x+40>0)./40);
end

function pattern_plot(theta,E1,style)
    eps = 10^-12;
    polarplot(theta,trunc(20*log10(eps+normalise(abs(E1)))),style)
    thetalim([min(theta)*180/pi max(theta)*180/pi]);               % only 0–180°
    rlim([0 1])
    rticks([0 0.25 0.5 0.75 0.99])
    rticklabels([-40 -30 -20 -10 0]);         % dB rings

    thetaticks(min(theta)*180/pi:30:max(theta)*180/pi);
    ax = gca;
    ax.ThetaZeroLocation = 'top';    % 0° at the top
    ax.ThetaDir = 'clockwise';       % match antenna books
end
```